

OAUTHC FAI External Validation — Microsoft Access Schema & Forms Specification

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Companion to `data_dictionary.md` and `oatuhc_fai_validation_protocol.qmd`.

Use this document to assemble the data-entry database in **Microsoft Access**

(2016 or later) on a Windows PC.

0. Overview

The database has a **two-table relational design**:

- ``tbl_Patient`` — one record per patient (demographics only, no identifiers)
- ``tbl_EarRecord`` — one record per ear (left/right), linked to patient via ``patient_id``

This mirrors the analysis file convention (one row per ear; two rows share a `patient_id`). Lookup tables provide dropdown values for constrained fields.

`tbl_Patient 1 ■■■■ ∞ tbl_EarRecord`

1. Tables

1.1 `tbl_Patient`

Field	Data type	Size/Format	Required	Notes
<code>patient_id</code>	AutoNumber	Long Integer	—	Primary key (assigned by Access)
<code>age_years</code>	Number	Integer (Byte)	Yes	Validation rule 18–110
<code>sex</code>	Number	Integer (Byte)	No	1 = male, 2 = female; blank = not documented
<code>created_at</code>	Date/Time	General Date	Yes	Default <code>`=Now()`</code> (audit)
<code>created_by</code>	Text	50	No	Data-entry operator (audit)

Table validation rule (sex): `[sex] Is Null Or [sex] In (1,2)`

1.2 `tbl_EarRecord`

Field	Data type	Size/Format	Required	Notes
<code>study_id</code>	Number	Long Integer	Yes	Sequential per ear record, unique
<code>patient_id</code>	Number	Long Integer	Yes	FK → <code>tbl_Patient.patient_id</code>
<code>ear</code>	Text	5	Yes	<code>`left` / `right`</code> (lookup)
<code>test_month_year</code>	Text	12	Yes	e.g. <code>`Mar 2023`</code> — month-year only, never day
<code>th_250</code>	Number	Single	No	dB HL, −10..120
<code>th_500</code>	Number	Single	No	dB HL
<code>th_1k</code>	Number	Single	No	dB HL
<code>th_2k</code>	Number	Single	No	dB HL
<code>th_3k</code>	Number	Single	No	dB HL
<code>th_4k</code>	Number	Single	No	dB HL
<code>th_6k</code>	Number	Single	No	dB HL
<code>th_8k</code>	Number	Single	No	dB HL
<code>bc_500</code>	Number	Single	No	Bone conduction, dB HL (optional)
<code>bc_1k</code>	Number	Single	No	
<code>bc_2k</code>	Number	Single	No	

Field	Data type	Size/Format	Required	Notes
bc_4k	Number	Single	No	
diagnosis_category	Text	20	No	Lookup `tbl_Lookup_Diagnosis`
consultant_grade	Text	20	No	Lookup `tbl_Lookup_Grade`; only where documented
documented_shape	Text	12	No	Lookup `tbl_Lookup_Shape`; only where documented
ear_included	Number	Integer (Byte)	No	1 = eligible, 0 = excluded
exclusion_reason	Text	100	No	Blank if included
created_at	Date/Time	General Date	Yes	Default `=Now()`

Indexes: unique index on study_id; index on patient_id;

unique composite index on (patient_id, ear) to prevent duplicate rows.

Field validation rules (thresholds):

`>= -10 And <= 120`

applied to every th_* and bc_* field.

Table validation rule (ear_included):

`[ear_included] Is Null Or [ear_included] In (0,1)`

Table validation rule (diagnosis):

`[diagnosis_category] Is Null Or [diagnosis_category] In ('csoM', 'otosclerosis', 'nihil', 'presbycusis', 'ssnihil', 'ototoxicity', 'meniere', 'other', 'not_documented')`

1.3 Lookup tables

tbl_Lookup_Diagnosis

Value	Display
csoM	Chronic suppurative otitis media
otosclerosis	Otosclerosis
nihil	Noise-induced hearing loss
presbycusis	Presbycusis
ssnihil	Sudden sensorineural hearing loss
ototoxicity	Ototoxicity
meniere	Meniere's disease
other	Other
not_documented	Not documented

tbl_Lookup_Grade

Value	Display
normal	Normal
mild	Mild
moderate	Moderate
moderately_severe	Moderately severe
severe	Severe
profound	Profound

tbl_Lookup_Shape

Value	Display
flat	Flat

Value	Display
sloping	Sloping
notched	Notched
rising	Rising

2. Relationships

Access → **Database Tools** → **Relationships**:

- `tbl\_Patient.patient\_id` (1) ■ `tbl\_EarRecord.patient\_id` (∞)

- **Enforce Referential Integrity: YES**

- **Cascade Update: YES** (patient_id never changes in practice, but harmless)

- **Cascade Delete: YES** (removing a patient removes both ear rows — desirable at data-entry stage)

3. Forms

3.1 `frm\_PatientEntry` (main form — single patient)

Purpose: capture demographics, then open the ear-record subform.

Controls:

Control	Bound to	Type	Notes
txtAge	tbl_Patient.age_years	Text box	Input mask `00`; BeforeUpdate validation
cboSex	tbl_Patient.sex	Combo box	RowSource `1;2` or lookup; blank allowed
txtCreatedBy	tbl_Patient.created_by	Text box	Optional operator initials

Form events:

- `Form\_BeforeUpdate` — validate `age\_years` 18–110:

```
Private Sub Form_BeforeUpdate(Cancel As Integer)
    If Not IsNull(Me.txtAge) Then
        If Me.txtAge < 18 Or Me.txtAge > 110 Then
            MsgBox "Age must be 18-110.", vbExclamation, "Validation"
            Cancel = True
            Me.txtAge.SetFocus
        End If
    End If
End Sub
```

- `Form\_AfterUpdate` — auto-open the ear-record form for this patient:

```
Private Sub Form_AfterUpdate()
    DoCmd.OpenForm "frm_EarRecord", , , "patient_id=" & Me.patient_id
End Sub
```

3.2 `frm\_EarRecord` (data-entry form — one ear per record)

Purpose: capture all audiogram fields for one ear of the current patient.

Controls (bound to tbl_EarRecord):

Section	Control	Bound to	Type/Notes
Header	txtStudyID	study_id	Read-only or auto-assigned (see 3.3)
	txtPatientID	patient_id	Read-only (set from main form)
	cboEar	ear	Combo `left;right`
	txtMonthYear	test_month_year	Text, input mask `LLL\ 0000` (e.g. Mar 2023)
Air thresholds	txtTh250 .. txtTh8k	th_250 .. th_8k	8 text boxes, tab order left→right
Bone conduction	txtBc500 .. txtBc4k	bc_500 .. bc_4k	4 text boxes
Clinical	cboDiag	diagnosis_category	Combo from tbl_Lookup_Diagnosis
	cboGrade	consultant_grade	Combo from tbl_Lookup_Grade

Section	Control	Bound to	Type/Notes
	cboShape	documented_shape	Combo from tbl_Lookup_Shape
QA	cboIncluded	ear_included	Combo `1;0`
	txtExclusion	exclusion_reason	Text box; enabled only if ear_included=0
Footer	lblPTA	(unbound)	Live PTA-4 display (see 3.4)
	lblMono	(unbound)	Live monotonicity warning (see 3.4)

Key events:

- `cboEar\_AfterUpdate` — prevent duplicate (patient_id, ear):

```
Private Sub cboEar_AfterUpdate()
    If Not IsNull(Me.patient_id) And Not IsNull(Me.cboEar) Then
        If DCount("'", "tbl_EarRecord",
            "patient_id=" & Me.patient_id & " And ear='" & Me.cboEar & "'") > 0 Then
            MsgBox "This ear already has a record for this patient.", vbExclamation
            Me.cboEar.Undo
        End If
    End If
End Sub
```

- Threshold `BeforeUpdate` (all 12 threshold boxes) — range check:

```
Private Sub txtTh250_BeforeUpdate(Cancel As Integer)
    If Not IsNull(Me.txtTh250) Then
        If Me.txtTh250 < -10 Or Me.txtTh250 > 120 Then
            MsgBox "Threshold must be -10 to 120 dB HL.", vbExclamation
            Cancel = True
        End If
    End If
End Sub
```

(Repeat for txtTh500 ... txtBc4k; or use the field-level Validation Rule instead — simpler.)

- `txtExclusion\_GotFocus` — require ear_included=0:

```
Private Sub txtExclusion_GotFocus()
    If IsNull(Me.cboIncluded) Or Me.cboIncluded <> 0 Then
        MsgBox "Set 'ear_included' to 0 before entering an exclusion reason.", _
            vbInformation
        Me.cboIncluded.SetFocus
    End If
End Sub
```

3.3 Auto-assigning study_id

Use a BeforeInsert event on frm_EarRecord:

```
Private Sub Form_BeforeInsert(Cancel As Integer)
    Me.txtStudyID = Nz(DMax("study_id", "tbl_EarRecord"), 0) + 1
End Sub
```

This guarantees sequential, unique study_id per ear record.

3.4 Live QA displays (footer labels)

Update on every threshold change (AfterUpdate of each threshold box or via

Form_AfterUpdate):

```
Private Sub UpdateQA()
    Dim pta As Variant, v As Variant, maxJump As Double, i As Integer
    Dim arr(1 To 8) As Variant, freqs(1 To 8) As String
    freqs(1) = "txtTh250": freqs(2) = "txtTh500": freqs(3) = "txtTh1k"
    freqs(4) = "txtTh2k": freqs(5) = "txtTh3k": freqs(6) = "txtTh4k"
    freqs(7) = "txtTh6k": freqs(8) = "txtTh8k"

    ' PTA-4 = mean(500, 1k, 2k, 4k)
    pta = Nz(Me.txtTh500, 0) + Nz(Me.txtTh1k, 0) + Nz(Me.txtTh2k, 0) + Nz(Me.txtTh4k, 0)
    If Not IsNull(Me.txtTh500) And Not IsNull(Me.txtTh1k) And _
        Not IsNull(Me.txtTh2k) And Not IsNull(Me.txtTh4k) Then
        pta = pta / 4
        Me.lblPTA.Caption = "PTA-4 = " & Format(pta, "0.0") & " dB"
    Else
        Me.lblPTA.Caption = "PTA-4 = - (need 500,1k,2k,4k)"
    End If
End Sub
```

```

' Monotonicity: any adjacent jump > 40 dB
For i = 1 To 8
    arr(i) = Nz(Controls(freqs(i)), Null)
Next i
maxJump = 0
For i = 1 To 7
    If Not IsNull(arr(i)) And Not IsNull(arr(i + 1)) Then
        If Abs(arr(i) - arr(i + 1)) > maxJump Then maxJump = Abs(arr(i) - arr(i + 1))
    End If
Next i
If maxJump > 40 Then
    Me.lblMono.Caption = "■ Adjacent jump " & Format(maxJump, "0") & " dB > 40 — REVIEW"
    Me.lblMono.ForeColor = RGB(192, 0, 0)
Else
    Me.lblMono.Caption = "Monotonicity OK"
    Me.lblMono.ForeColor = RGB(0, 128, 0)
End If
End Sub

```

Call UpdateQA from the AfterUpdate event of every threshold control.

4. Queries (analysis prep)

4.1 `qry\_DataEntryQA` — batch validation check

```

SELECT e.study_id, e.patient_id, e.ear, p.age_years, p.sex,
    e.th_500, e.th_1k, e.th_2k, e.th_4k,
    e.ear_included, e.exclusion_reason,
    (e.th_500 + e.th_1k + e.th_2k + e.th_4k) / 4 AS pta4,
    IIf(IsNull(e.th_250) Or IsNull(e.th_500), 0, Abs(e.th_250 - e.th_500)) AS j_250_500,
    IIf(IsNull(e.th_500) Or IsNull(e.th_1k), 0, Abs(e.th_500 - e.th_1k)) AS j_500_1k,
    IIf(IsNull(e.th_1k) Or IsNull(e.th_2k), 0, Abs(e.th_1k - e.th_2k)) AS j_1k_2k,
    IIf(IsNull(e.th_2k) Or IsNull(e.th_3k), 0, Abs(e.th_2k - e.th_3k)) AS j_2k_3k,
    IIf(IsNull(e.th_3k) Or IsNull(e.th_4k), 0, Abs(e.th_3k - e.th_4k)) AS j_3k_4k,
    IIf(IsNull(e.th_4k) Or IsNull(e.th_6k), 0, Abs(e.th_4k - e.th_6k)) AS j_4k_6k,
    IIf(IsNull(e.th_6k) Or IsNull(e.th_8k), 0, Abs(e.th_6k - e.th_8k)) AS j_6k_8k,
    (IIf(IsNull(e.th_250) Or IsNull(e.th_500), 0, Abs(e.th_250 - e.th_500)) +
    IIf(IsNull(e.th_500) Or IsNull(e.th_1k), 0, Abs(e.th_500 - e.th_1k)) +
    IIf(IsNull(e.th_1k) Or IsNull(e.th_2k), 0, Abs(e.th_1k - e.th_2k)) +
    IIf(IsNull(e.th_2k) Or IsNull(e.th_3k), 0, Abs(e.th_2k - e.th_3k)) +
    IIf(IsNull(e.th_3k) Or IsNull(e.th_4k), 0, Abs(e.th_3k - e.th_4k)) +
    IIf(IsNull(e.th_4k) Or IsNull(e.th_6k), 0, Abs(e.th_4k - e.th_6k)) +
    IIf(IsNull(e.th_6k) Or IsNull(e.th_8k), 0, Abs(e.th_6k - e.th_8k))) AS max_jump
FROM tbl_EarRecord AS e
INNER JOIN tbl_Patient AS p ON e.patient_id = p.patient_id
WHERE e.ear_included = 1;

```

Any row where max_jump > 40 is a **monotonicity flag** — review, do not auto-correct.

4.2 `qry\_ExportFAI` — clean export for the FAI pipeline

```

SELECT e.study_id, e.patient_id, e.ear, e.test_month_year,
    p.age_years, p.sex,
    e.th_250, e.th_500, e.th_1k, e.th_2k, e.th_3k, e.th_4k, e.th_6k, e.th_8k,
    e.bc_500, e.bc_1k, e.bc_2k, e.bc_4k,
    e.diagnosis_category, e.consultant_grade, e.documented_shape,
    e.ear_included, e.exclusion_reason
FROM tbl_EarRecord AS e
INNER JOIN tbl_Patient AS p ON e.patient_id = p.patient_id
WHERE e.ear_included = 1
ORDER BY e.study_id;

```

5. Macros / VBA — batch operations

5.1 Export to CSV (one click)

Module mod_Export:

```

Public Sub ExportFAI_CSV()
    Dim qd As QueryDef, rst As DAO.Recordset, fso As Object, ts As Object
    Dim path As String, i As Integer, line As String

    path = CurrentProject.Path & "\fai_export_" & Format(Now, "yyyymmdd_hhnn") & ".csv"
    Set fso = CreateObject("Scripting.FileSystemObject")

```

```

Set ts = fso.CreateTextFile(path, True, True)

Set qd = CurrentDb.QueryDefs("qry_ExportFAI")
Set rst = qd.OpenRecordset()

' Header
line = ""
For i = 0 To rst.Fields.Count - 1
    line = line & IIf(i > 0, ",", "") & rst.Fields(i).Name
Next i
ts.WriteLine line

' Rows – blank as empty (no NA/./0)
Do While Not rst.EOF
    line = ""
    For i = 0 To rst.Fields.Count - 1
        If IsNull(rst.Fields(i).Value) Then
            line = line & IIf(i > 0, ",", "") & ""
        Else
            line = line & IIf(i > 0, ",", "") & CStr(rst.Fields(i).Value)
        End If
    Next i
    ts.WriteLine line
    rst.MoveNext
Loop

ts.Close
rst.Close
MsgBox "Exported: " & path, vbInformation
End Sub

```

5.2 Auto-increment check (QA)

```

Public Function MissingStudyIDs() As Long
    ' Returns count of duplicate/missing study_ids for QA log
    Dim cnt As Long
    cnt = DCount("*", "qry_DuplicateStudyID")
    Debug.Print "Duplicate study_ids: " & cnt
    MissingStudyIDs = cnt
End Function

```

(SQL for qry_DuplicateStudyID:

```

SELECT study_id, COUNT(*) AS n
FROM tbl_EarRecord
GROUP BY study_id
HAVING COUNT(*) > 1;

```

)

6. Access → FAI pipeline handoff

1. Run qry_DataEntryQA after every batch of 20 — resolve flags.
2. Run ExportFAI_CSV → fai_export_YYYYMMDD_HHNN.csv.
3. Feed CSV to the Python validation script (repo-side):

```
python scripts/validate_oatuhc.py --input fai_export_YYYYMMDD_HHNN.csv
```

4. The script computes frozen-FAI fai_score, fai_label, fai_config, membership degrees, WHO grade, and the borderline flag — same contract as data_dictionary.md derived variables.

7. Security & hygiene

- ****No PHI.**** The schema collects no names, hospital numbers, addresses, phones or emails. patient_id is an arbitrary sequential number.
- Password-protect the database if multiple operators share a machine (File → Encrypt with Password).
- Keep a dated backup of the .accdb before each export batch.
- Store the .accdb on a network drive with per-user NTFS permissions if multi-user concurrent entry is needed; otherwise single-user on the

entry machine and nightly backup.

- **Do not** store `created\_by` as free text if operator names are sensitive — use a user code (initials).

8. Build checklist (Access steps in order)

- [] 1. Create tables: `tbl\_Patient`, `tbl\_EarRecord`, 3 lookup tables
- [] 2. Set field data types, validation rules, indexes per §1
- [] 3. Define relationships with referential integrity (§2)
- [] 4. Build `frm\_PatientEntry` with BeforeUpdate/AfterUpdate VBA (§3.1)
- [] 5. Build `frm\_EarRecord` with all controls + events (§3.2–3.4)
- [] 6. Create queries `qry\_DataEntryQA`, `qry\_ExportFAI`, `qry\_DuplicateStudyID` (§4)
- [] 7. Add module `mod\_Export` + ExportFAI_CSV (§5.1)
- [] 8. Test: enter 2 rows for one patient (L/R), export CSV, verify blank handling
- [] 9. Back up .accdb; hand to data-entry operators with the Instructions sheet